



Vapor-liquid equilibrium data for the binary system isopropanol+water at 60 kPa and 80 kPa

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ABSTRACT

In this work, experimental Vapor-Liquid Equilibrium (VLE) data for the Isopropanol + Water system under vacuum conditions at 60 kPa and at 80 kPa, for which no experimental evidences have been found in the literature, are reported. The experimental investigation of the isobaric equilibrium has been carried out by GASP at the Process Thermodynamics laboratory (PT lab) of Politecnico di Milano by employing an all-glass dynamic recirculation still. The collected experimental data, together with those already available in the literature at other conditions, have been used for the evaluation of the thermodynamic model that best fits the data. Four different models have been considered and their performances have been checked through the calculation of the Absolute Average Deviation (AAD%). The lowest AAD% has been found for the UNIFAC model only for isobaric data, not for the isothermal points. The regression tool available in Aspen Plus® V14 has been used to fit the binary interaction parameters of the NRTL activity model.

1. Introduction

2-Propanol (also known as IsoPropyl Alcohol, IPA) is a key chemical with a growing market, projected to reach USD 5.1 billion by 2030 with a Composite Annual Growth Rate (CAGR) of 7.8 % [1]. Its birth goes back to the latter part of the First World War [2] and actually IPA is claimed to be the first petrochemical produced. At that time it was manufactured following the Ellis process, which is based on the reaction of propylene with sulphuric acid to form isopropyl sulfates, that are then hydrolysed to isopropanol [3]. Nevertheless its industrial relevance is still outstanding and that is why the scientific research is still focusing on this compound. In 2021, during the European Petrochemical Association Conference (EPCA), BASF introduced a new range of catalysts to produce IPA more efficiently believing that it would become a more and more crucial ingredient in hand sanitizers, a much needed consumer product during the Covid-19 pandemic. In addition, its application in the research field has not to be underestimated. The IPA cation is a perfect system to compare the alkyl vs. alkane elimination kinetics and energetics. Indeed, the two processes usually struggle at the same bond cleavage side. However, since IPA contains two identical methyl groups and has three different sites with abstractable hydrogens, it can undergo different dissociation pathways, whose thermodynamic analysis can

provide important insights about the mentioned elimination processes [4]. IPA also complies to today's need to find out new alternatives to fossil fuels [5] and it is already used as gasoline and diesel additive [6,7]. As alcohol, it is considered to be an alternative transportation fuel because it could be used in existing engine with minor hardware modifications [8]. Its Research Octane Number (RON 129) [9] is higher than the one of gasoline, meaning that it can endure higher compression ratios before the engine starts knocking, thus giving to the engine the capacity to deliver more power efficiently and economically. Even though the research on fuel substitutes originally focused on ethanol, it turns out that this primary alcohol is not the ideal biofuel. Its corrosivity and hygroscopicity make it incompatible with the existing fuel storage and distribution infrastructure and moreover ethanol contains only the 70 % of the energy content of gasoline [6]. Isopropanol has an energy density of 23.9 MJ/L, somewhat less than the one of gasoline (32 (MJ/L)), and low water solubility when compared to ethanol. This latter property is of interest, considering that the water content in fuels has always been a problem, because of its ability to promote severe damage on fuel-injection equipment, to promote emulsions in the oil and rusting of the ferrous components [10,11]. The fact that isopropanol can be produced exploiting genetically engineered microorganisms as the *Cupriavidus necator*, even meeting the requirement for engineering a

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microorganism for industrial metabolite production [9], makes IPA appealing as a fourth-generation biodiesel. Moreover, though the research on the fourth-generation biodiesel is still in the early stage, the Fatty Acid Propyl Esters (FAPRE) are being attractive to researchers, either alone or in binary or ternary blends with mineral diesel used as fuel in diesel engine, and lots of papers have been written in the last twenty years demonstrating the positive effect on the low-temperature properties of the biodiesels when the propanol isomer is used for their synthesis [12,13,14,15,16]. With respect to the traditional Fatty Acid Methyl Esters (FAMES) or the Fatty Acid Alkyl Esters (FAAEs), the FAPREs have a higher cetane number, as demonstrated in a paper by Klopfenstein [17] and also, considering the alcohol moiety of the FAAEs, it was shown that as the alcohol chain increases, other fuel properties, such as the heat of combustion and the reduction of NO_x emissions, improve as well [18]. Finally Hellier [19] demonstrated that the commercial FAME can be easily replaced by biodiesel made of isopropanol or propanol because of the negligible effect of the chain length or the structural branching on the ignition delay, promoting the use of IPA also for this utilization. Nowadays, isopropanol is mainly used as intermediate for chemicals production (as acetone, hydrogen peroxide, etc. [20]) and also as a solvent in the pharmaceutical industry [21]. High concentration of IPA in water is also widely used as cleaning agent in modern electronic industries [22], like the semiconductor industries, where more than thousand tons of high purity IPA per year can be exploited for cleaning purposes [23]. The relative waste solvent stream contains mainly Isopropyl Alcohol and water [24] and needs to be recovered and recycled back to the process because of the environmental and economic constraints [22]. In addition, the IPA and water mixture is the result of the IPA synthesis according to the propylene hydration, which is still the most important way to prepare IPA [25]. Therefore, the necessity to separate the IPA and water mixture rises. However, due to the formation of a minimum boiling azeotrope (at about 87.5 % wt. of IPA at 101.3 kPa), the separation can not be performed by means of a standard distillation column and special distillation techniques for the IPA dehydration have received widespread attention, such as azeotropic distillation [26,27,28,29,30], extractive distillation exploiting different entrainers as Dymethyl Sulfoxide (DMSO) [24,31], Choline-based deep eutectic solvents [32], ionic liquids as 1-Ethyl-3-methylimidazolium acetate ([EMIM][OAc]) [33] or 1-Ethyl-3-methylimidazolium dicyanamide ([emim][N(CN)₂]) [34] and pressure swing distillation [35] to name a few. Some of these components could degrade at a temperature close to their boiling point at atmospheric pressure, as for instance DMSO, degrading at temperatures higher than 190 °C. The processes employing these compounds are therefore characterized by sections of the plant at pressure lower than the atmospheric one, to decrease the operating temperature. To simulate and design such processes, reliable phase behavior data and representation thereof are required. This work has been carried out at the Process Thermodynamics laboratory (PT lab) of Politecnico di Milano, held by the research group GASP. The aim of this work is to provide new measurements of Vapor–Liquid Equilibrium (VLE) at 80 kPa, at which no data are available in the literature though being of interest, and at 60 kPa, conditions for which only another source [36] has been found, and to compare the ability of both Non Random Two-Liquids (NRTL) [37] and Dortmund modified UNIFAC model [38,39,40,41] to describe the experimental data.

1.1. VLE data available in the literature.

The literature provides a large amount of VLE data for the 2-Propanol + Water system. The complete data sets reporting pressure, temperature and composition of both the liquid and vapour phases are listed in Table 1 and in Table 2. It can be observed that most of the available literature works focused on the collection of VLE data at about atmospheric pressure or at higher pressure (except Wilson and Simons [42], Marzal et al. [36] and Zhao et al. [43]), including the data previously collected at PT lab at pressures between 150 kPa and 300 kPa [44]. As

Table 1

Isobaric experimental vapor–liquid equilibrium data for the 2-Propanol + Water system.

Source	Pressure (kPa)	No. experimental points	T _{min} (K)	T _{max} (K)
Wilson and Simons, 1952 [42]	12.666	24	308.93	323.86
Wilson and Simons, 1952 [42]	25.331	24	322.38	338.44
Marzal et al., 1996 [36]	30	25	325.33	342.33
Wilson and Simons, 1952 [42]	50.663	25	337.06	354.83
Marzal et al., 1996 [36]	60	26	340.52	359.14
Zhao et al., 1986 [43]	85.3	14	349.53	351.63
Ratkovics and Sayed, 1970 [45]	98.925	8	352.53	367.13
Marzal et al., 1996 [36]	100	27	352.65	372.78
Lebo, 1921 [46]	100.658	40	353.25	371.55
Brunjes and Bogart, 1943 [47]	101	28	353.18	372.02
Verhoeve, 1968 [48]	101	9	353.25	356.95
Kohoutová et al., 1970 [49]	101	20	353.23	364.63
Gironi and Lamberti, 1995 [50]	101	32	353.37	373.15
Dunlop, 1948 [51]	101.32	26	353.12	372.02
Verhoeve, 1970 [52]	101.3	13	353.26	359.53
Rajendran et al., 1991 [53]	101.3	14	353.15	359.05
Gu and Hou, 2000 [54]	101.3	10	353.15	357.07
Sevgili and Senol, 2006 [55]	101.3	17	353.16	373.15
Lin and Tu, 2014 [56]	101.3	17	353.22	373.15
Zhang et al., 2016 [21]	101.3	14	353.32	365.42
Gupta et al., 2017 [57]	101.3	15	353.5	373.2
Choffé and Asselineay, 1956 [58]	101.32	11	353.52	373.12
Dobroserdov, 1959 [59]	101.32	14	353.31	366.32
Yorizane et al., 1967 [60]	101.32	24	353.15	373.15
Kojima et al., 1968 [61]	101.32	21	353.4	373.15
Kojima et al., 1968 [62]	101.32	20	353.61	373.15
Ohe et al., 1969 [63]	101.32	15	353.45	365.35
Udovenko et al., 1973 [64]	101.32	20	353.53	373.12
Arce et al., 2003 [65]	101.32	27	353.28	371.95
Li et al., 2008 [66]	101.32	11	353.20	357.15
Li et al., 2009 [67]	101.32	12	353.34	373.15
Wilson and Simons, 1952 [42]	101.325	26	353.3	373.15
Verhoeve and De Schepper, 1973 [68]	101.325	11	353.26	359.53
Moioli et al., 2021 [44]	103.5	25	353.90	373.69
Moioli et al., 2021 [44]	150	22	363.48	381.53
Moioli et al., 2021 [44]	200	24	371.33	390.50
Moioli et al., 2021 [44]	250	24	377.83	397.06
Moioli et al., 2021 [44]	300	18	383.31	403.28
Wilson and Simons, 1952 [42]	411.566	17	393.55	417.76

for low pressure data, only Marzal et al. [36] collected some points at 60 kPa in 1996 and no data have been found at 80 kPa. The closer value of pressure has been considered by Zhao et al. [43], who presented experimental points at a pressure of about 85 kPa.

2. Materials and methods

2.1. Materials

All the characteristics of the chemicals used are summarized in Table 3. 2-Propanol (CAS 67–63-0) has been supplied by Sigma-Aldrich with a declared purity of at least 99.5 % wt. (the purity reported in Table 3 is the one measured at PT lab, confirming it is higher than 99.5 % wt.), a density of 785 kg/m³ and a molecular weight of 60.1 g/mol.

Table 2

Isothermal experimental vapor–liquid equilibrium data for the 2-Propanol + Water system.

Source	Temperature (K)	No. experimental points	P _{min} (kPa)	P _{max} (kPa)
Sazonov, 1986 [69]	298.15	14	3.16	6.65
Udovenko and Mazanko, 1967 [70]	303.15	18	4.28	8.92
Sada and Morisue, 1975 [71]	308.16	9	8.59	11.60
Udovenko and Mazanko, 1967 [70]	318.15	18	9.65	20.41
Sada and Morisue, 1975 [71]	318.21	6	15.25	19.99
Sada and Morisue, 1975 [71]	328.18	5	25.72	33.12
Udovenko and Mazanko, 1967 [70]	333.15	18	19.99	42.58
Sada and Morisue, 1975 [71]	338.19	5	41.64	52.86
Sada and Morisue, 1975 [71]	348.12	3	64.27	80.05
Wu et al., 1988 [72]	353.15	21	47.49	100.34
Barr-David and Dodge, 1959 [73]	423.15	20	517.11	917.00
Barr-David and Dodge, 1959 [73]	473.15	19	1847.80	2806.17
Barr-David and Dodge, 1959 [73]	523.15	16	4309.23	6618.97
Shimoyama et al. [74]	548	3	6490	7990
Barr-David and Dodge, 1959 [73]	548.15	18	6901.65	9307.93
Shimoyama et al. [74]	573	6	9560	11,480
Barr-David and Dodge, 1959 [73]	573.15	6	8894.24	12348.52

Demineralized water (ISO 3696 Q3, ASTM D 1193 TYPE 4) has been employed. Gaseous Nitrogen (>99.99 % mol.) has been used for the pressure control of the VLE unit (Fischer® Labodest® VLE 602) and as a carrier for the gas chromatograph (GC). The purity of IPA has been checked by GC analysis at PT lab.

The density of the pure substances has also been measured experimentally in the PT lab by using the Mettler Toledo Densito Density2Go, characterized by an accuracy on the measurement of density equal to 1 kg/m³ and on the measurement of temperature equal to 0.2 °C, with three measurements for each substance. The densimeter had been previously calibrated on water standards with the default test “water”. The density of IPA, measured at 19.37 °C, is 786 kg/m³ and the density of water, measured at 18.53 °C, is 999 kg/m³. For the density of IPA, Kireev

et al. [75] report a value of 786.178 kg/m³ at 19.94 °C and Vostrikova et al. [76] report a value of 786 kg/m³ at the same temperature. For the density of water, a value equal to 998.6 kg/m³ has been determined by Kurtz et al. [77] at 19.84 °C and equal to 999 kg/m³ by Vaidya and Naik [78]. The values of density obtained at PT lab are in agreement with the ones reported in the literature.

2.2. Experimental apparatus

An all-glass dynamic recirculating still (Fischer® Labodest® VLE 602, Fig. 1) [75] coupled to a vacuum pump has been used to collect the VLE experimental data of the mixture of interest [76]. The unit is provided with a concentric electrical immersion heater (1), which evaporates part of the liquid mixture fed to the unit. Within the Cottrell pump (2) a very intensive heat and mass-transfer occurs between the rising vapor and the boiling liquid, which are then directed to the separation chamber (3), where a Pt-100 resistance thermometer (8) (the accuracy and the resolution are ± 0.1 K and 0.01 K as indicated by the supplier) is used to measure the equilibrium temperature. Herein the vapor and liquid phases are separated. The vapor phase is condensed, and then all the phases return to the mixing chamber (7), where the mixture is homogenized by means of a magnetic stirrer and, then, fed back again to the immersion heater chamber. The unit can be operated under vacuum conditions up to 400 kPa and up to a temperature of 523.15 K (the accessories for the operation at high temperatures comprise the heating mantle (17), to compensate heat losses and to avoid stress in the glass, and the heating line (16) to avoid any blocking in the capillary tube). The heating element and the pressure are regulated with the digital controller, whose pressure transmitter has an accuracy of ± 100 Pa, as indicated by the supplier. The expanded uncertainties have, then, been calculated for each collected point.

2.3. Experimental procedure

The experiments have been carried out at PT lab of Politecnico di Milano with the procedure detailed in the following. The VLE unit is first connected to the vacuum pump, whose trap is placed into a Dewar filled with a 750 ml mixture of ice and alcohol (80 % vol. ice and 20 % vol. alcohol), in order to prevent the flowing of undesirable gases into the pump, and then the nitrogen supply valve is opened. The mixture (350 mL) is previously prepared under the hood in a flask with stopper. One third of it is loaded into the still after being mixed, the remaining two third are kept in a closed flask. By opening a PTFE valve at the feed vessel, the mixture is charged into the heating chamber so that the electric heating element (1) is completely submerged. Before starting the experiment, a check that the valves (12, 18, 13, 19) are open and the valves (14, 15) are closed is made. Then, the cooling water is switched on to allow it to flow through all the condensers (4, 5, 6) and, then, the magnetic stirrer is turned on at its maximum speed (and the liquid level in the heating chamber is checked again). The desired pressure value is set, the vacuum pump is switched on and once the unit reaches the pressure set point, the heating element is activated too. When the temperature detected by the Pt-100 (8) remains stable for at least 10 ÷ 15 min and 1–2 drops per second of condensate can be observed at the dripping tip (11), the thermodynamic equilibrium is considered to be

Table 3

Chemicals description.

Chemical name	Chemical formula	CAS number	Supplier	Purity	Analysis method
2-Propanol	C ₃ H ₈ O	67–63-0	Sigma-Aldrich	0.998 ^a	GC ^b performed at PT lab
Demineralized water	H ₂ O	7732–18-5	Idrochimica Srl	ASTM D 1193 TYPE 4	analysis performed by the supplier
Nitrogen	N ₂	7727–37-9	Sapio	>0.9999 ^c	analysis performed by the supplier

^a mass fraction.

^b Gas Chromatography.

^c mole fraction.

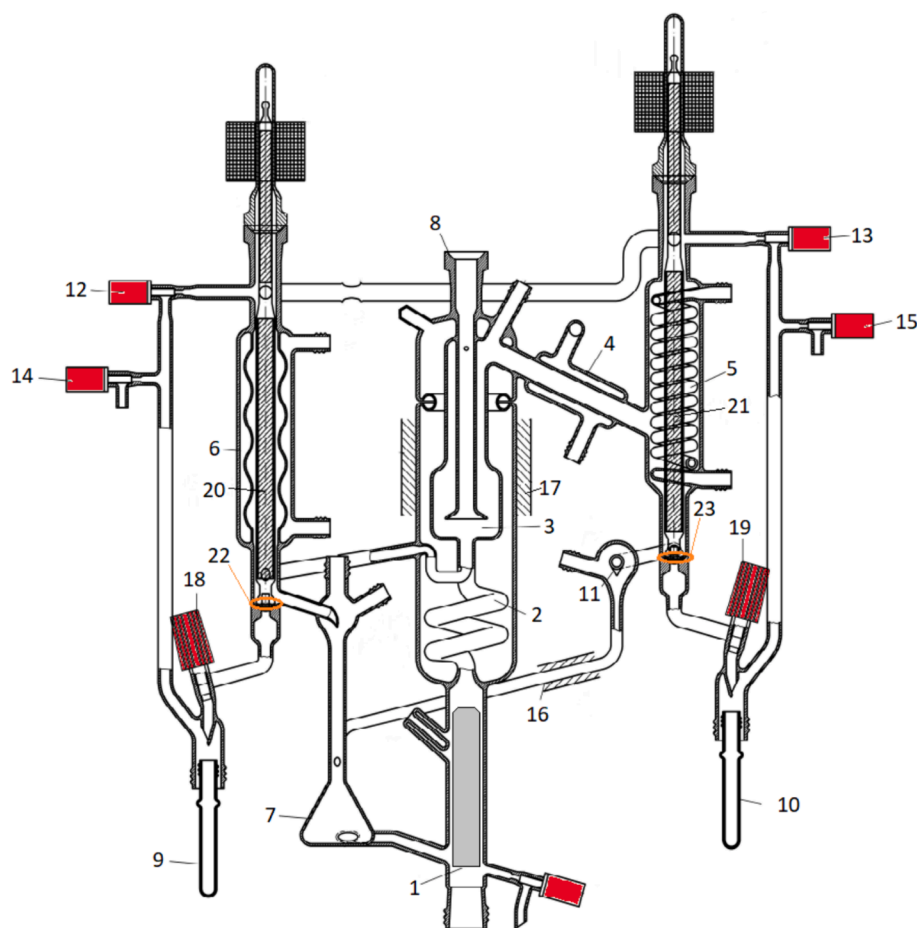


Fig. 1. Schematic representation of the experimental setup: 1, heater; 2, Cottrell-pump; 3, separation chamber; 4, condenser; 5, safety cooler vapor phase; 6, safety cooler liquid phase; 7, mixing chamber with PTFE stirrer bar; 8, Pt-100 vapor temperature; 9, receiver of the liquid sample; 10, receiver of the vapor sample; 11, dripping tip condensed vapor phase; 12, PTFE-valve for the pressure equilibrium to receiver No. 9; 13, PTFE-valve for the pressure equilibrium to receiver No. 10; 14, ventilation valve liquid sample; 15, ventilation valve vapor sample; 16, heating line; 17, heating mantle; 18, PTFE-valve for the liquid sample; 19, PTFE-valve for the vapor sample; 20, solenoid valve rod liquid phase; 21, solenoid valve rod vapor phase; 22, liquid accumulation point; 23, condensate accumulation point.

reached and the vapor sample, as condensate, and the liquid sample are collected at the same time. The analysis of the composition of the two phases is, then, performed in a GC (Agilent 7820A) equipped with a Thermal Conductivity Detector (TCD), the stationary phase being a HP-INNOWAX with a film thickness of 0.5 μm . The calibration of GC had been previously performed [44]. The unit is emptied and before filling it with the second portion of the already prepared mixture, a fictitious sampling is performed. Indeed some liquid and condensate may accumulate nearby the solenoid valves rods, at the (22) and (23) points, contaminating the liquid and vapor samples of the next test. Then, the overall procedure is repeated at least three times to reduce the uncertainties in the measurements and to ensure the repeatability of the experiment.

By preparing the amount of mixture (350 mL) able to carry out three single equilibrium measurements only once, the intrinsic error which may rise by preparing the same mixture (approximately 110 mL) three times is for sure lowered. In addition, by filling the storage tank of the unit with just the amount of mixture able to perform one equilibrium measurement, any possible stratification of the mixture within the tank is prevented and finally the overall composition of the mixture is not affected by the vapor and liquid sampling because, after the sampling, the unit is emptied and then refilled with another portion of the original mixture. The power to be provided to the unit is the other key parameter. In order to guarantee the stability of the temperature in the system, the power to be supplied to the heating element has to be increased as the amount of the high boiling component present in the mixture fed to

the unit increases.

2.4. Validation of the method

The unit had been employed in a previous work [44] for collecting vapor pressure data of pure water and IPA and the obtained points had been compared with the literature. Fig. 2 reports the collected points, detailed in [44], and the comparison with other literature sources.

The normal boiling point (at a pressure of 101.3 kPa) of IPA has been determined to be 81.9 $^{\circ}\text{C}$ ($U(T) = 0.03$ K) and the one for water to be 99.95 $^{\circ}\text{C}$ ($U(T) = 0.03$ K). The reliability of the experimental method employed in this work and of the unit has been verified by comparing the data collected within the PT lab with the available previously published results. This test had been performed considering the data of water and IPA vapor pressures between 31.07 kPa and 350 kPa and the VLE data for the 2-Propanol + Water system at atmospheric pressure [44].

Fig. 3 shows the agreement of the gathered points at 80 kPa to the trend highlighted in Moiola et al. [44]. The azeotropic mixture is characterized by a higher IPA mole fraction as the equilibrium pressure increases. A similar agreement has been obtained by comparing the isobaric points at 60 kPa collected in this work with the ones collected by Marzal et al. [36], though with some differences occurring for values of mole fractions close to 0.5, probably because of some instability issues in the boiling of the mixture. The azeotropic point obtained in this work is for a mole fraction of IPA in the liquid phase close to 0.6492 and is in

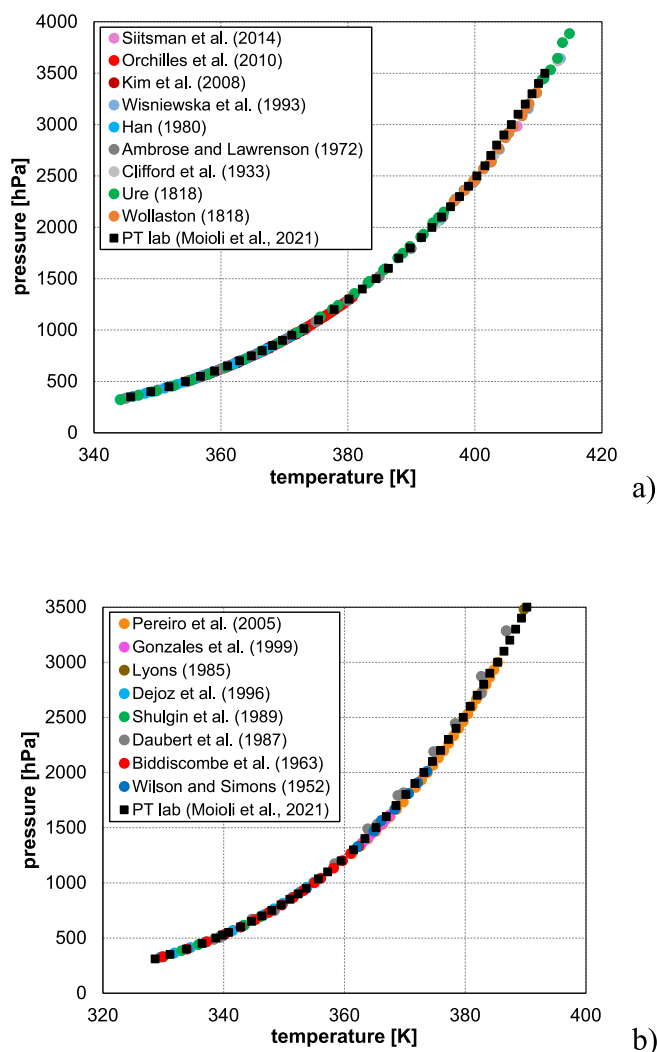


Fig. 2. Experimental data of vapor pressure of a) water and b) IPA collected at PT lab of Politecnico di Milano and comparison with data from the literature.

the range between the two values including the azeotropic point reported in Marzal et al. [36].

In addition, the thermodynamic consistency of all the obtained experimental data has been tested. Further details of the validation can be found in [44].

3. Experimental results

The experimental curves for the isobaric VLE measurements at 60 kPa and at 80 kPa are shown in Fig. 4 and in Fig. 5 and the single experimental points are reported in Table 4 and in Table 5 together with the average estimated uncertainties. These latter ones have been calculated following the procedure suggested by NIST (National Institute of Standards and Technology) in 1992, which are based on the approach recommended by the CIPM (International Committee for Weights and Measures) [77]. This work considers the expanded uncertainties, calculated with a 95 % level of confidence.

The provided expanded uncertainties have been estimated considering a coverage factor equal to 2, which ensures a 95 % level of confidence for the interval. Their averages have been calculated excluding the points for the pure components, for which the uncertainty in the mole fractions is 0.

The uncertainties associated with pressure deserves a dedicated discussion. The pressure is set at the beginning of each VLE

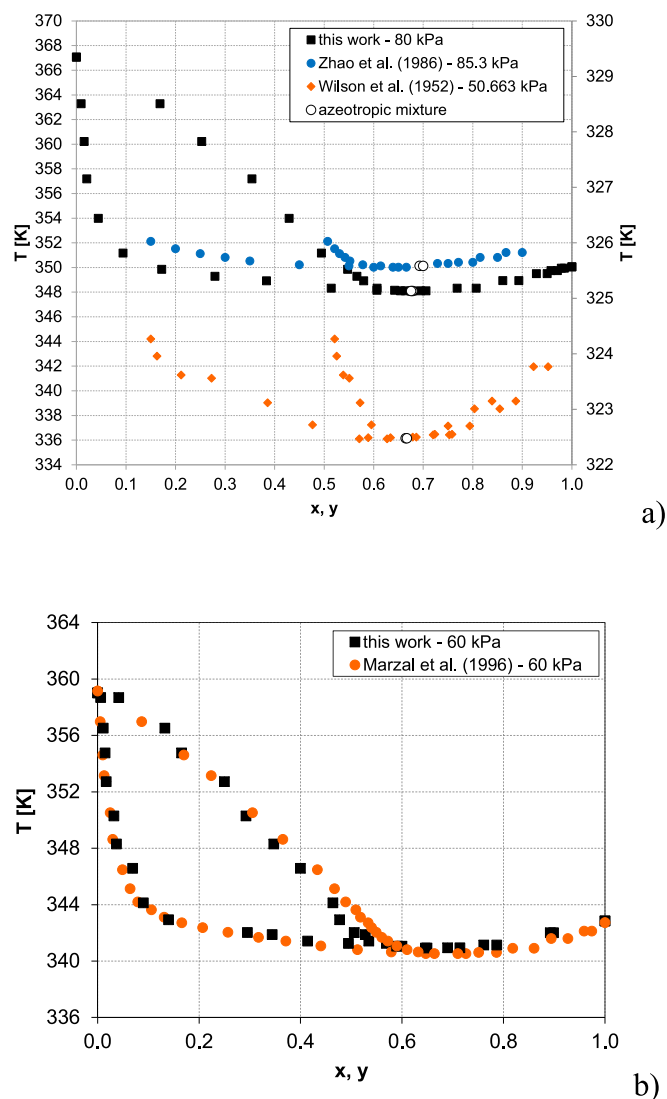


Fig. 3. Comparison of the VLE experimental points collected at PT lab of Politecnico di Milano for the 2-Propanol + Water system a) at 80 kPa and the ones by Wilson et al. (1952) at 50.663 kPa (plotted with respect to the secondary axis) and by Zhao et al. (1986) at 85.3 kPa (the yellow dots represent the azeotropic mixtures at the different pressures) and b) at 60 kPa and the ones by Marzal et al. [36].

measurement and its value is kept fixed by the pressure controller of the Fischer® Labodest® VLLE 602 unit, however being an on-off controller, small pressure fluctuations can occur. At a set pressure its oscillations have been measured at the beginning of each test and these values are the ones reported in the Table 4 and in Table 5 as pressure uncertainties, $U(\text{pressure})$.

By looking at Fig. 4 and Fig. 5 and considering the data reported in Table 4 and in Table 5 it is possible to observe that by increasing the pressure the azeotrope occurs at a higher concentration of 2-Propanol (at about 0.64 mol fraction of 2-Propanol at 60 kPa and between 0.65 and 0.66 mol fraction of 2-Propanol at 80 kPa). This is also confirmed by Fig. 6, which shows isobaric VLE data presented in this work and reported by literature papers at different pressures below 100 kPa (only the data sets that cover the entire composition range have been considered).

The thermodynamic consistency test has been performed, on the basis of the method by Redlich and Kister [78], for all the obtained experimental data. This test has been chosen to verify the thermodynamic consistency of the experimental data because of its wide use in the

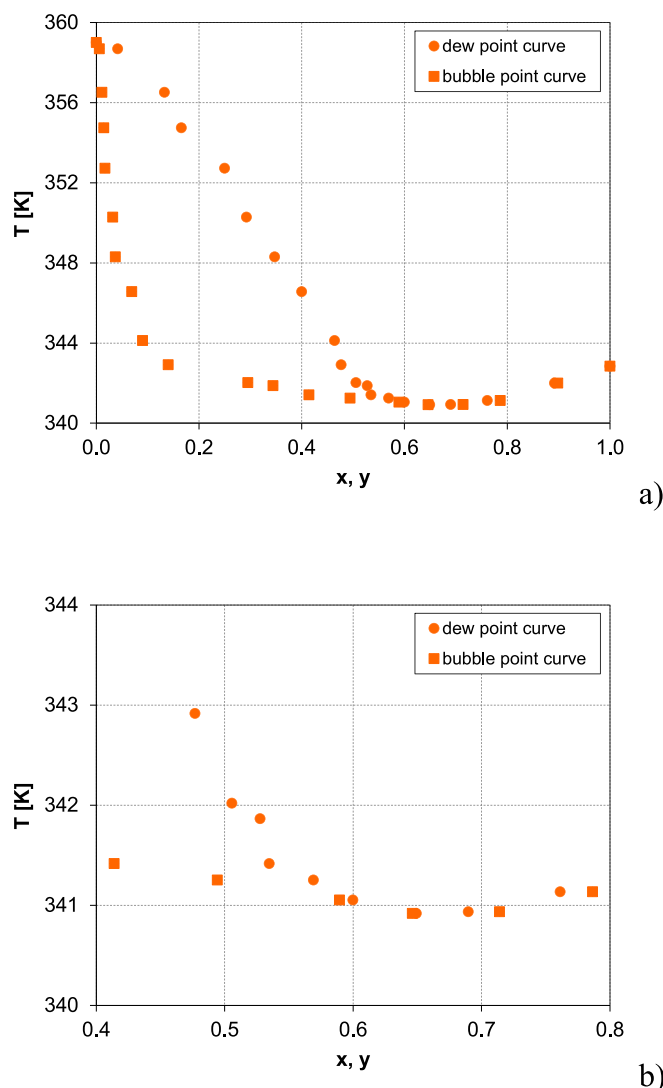


Fig. 4. Experimental VLE data collected at PT lab of Politecnico di Milano for the 2-Propanol + Water system at 60 kPa a) for a mole fraction of IPA ranging from 0 to 1 and b) for a mole fraction of IPA close to the azeotropic point.

literature [44] and because it does not require any thermodynamic model, differently from a point-to-point test, so the results do not vary. The results are reported in Table 6.

4. Thermodynamic modelling

The NRTL and Wilson models are generally preferred to describe the 2-Propanol + Water thermodynamic behavior [21,52,68,53,36,65,67,57,71,72]. Therefore, these two activity coefficient models have been considered in this work. In addition, another γ/φ model, the UNIFAC (Dortmund modified), and a φ/φ model, the PSRK, have been tested (also Shymoyama et al. [74] employed this Equation of State, even though the energy parameter of the original SRK had been modified to better represent the vapor pressures of polar substances [79]). Aspen Plus® V11.0, with its built-in model parameters, has been employed to evaluate the predicted variables, *i.e.* temperature and the mole fractions of IPA and water in the vapor phase, for the isobaric datasets, and pressure and the molar fractions of IPA and water in the vapor phase for the isothermal datasets.

Moreover, after performing the thermodynamic consistency test for the NRTL method the regression of new thermodynamic parameters has been accomplished considering all the experimental data collected at the

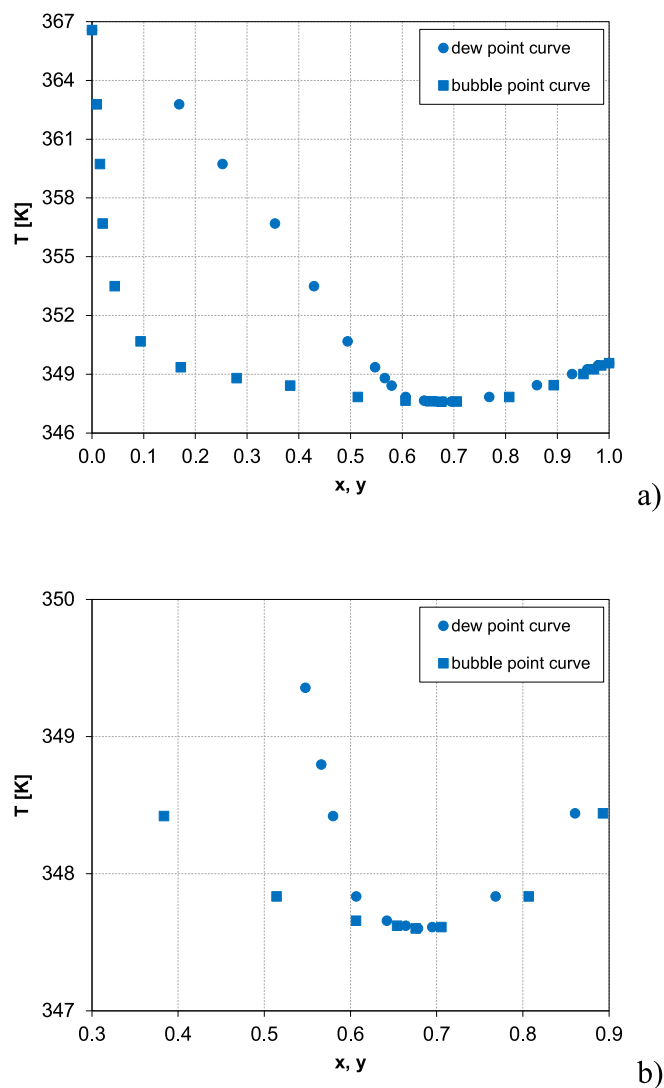


Fig. 5. Experimental VLE data collected at PT lab of Politecnico di Milano for the 2-Propanol + Water system at 80 kPa a) for a mole fraction of IPA ranging from 0 to 1 and b) for a mole fraction of IPA close to the azeotropic point.

PT lab and those available in the literature [21,46–74], both isothermal and isobaric data. For the sake of clarity, it is recalled here that the NRTL model calculates the activity coefficient according to Eq. (1):

$$\ln \gamma_i = \frac{\sum_j x_j \tau_{ji} G_{ji}}{\sum_k x_k G_{ki}} + \sum_j \frac{x_j G_{ij}}{\sum_k x_k G_{kj}} \left(\tau_{ij} - \frac{\sum_m x_m \tau_{mj} G_{mj}}{\sum_k x_k G_{kj}} \right) \quad (1)$$

where:

$$\begin{aligned} G_{ij} &= \exp(-\alpha_{ij} \tau_{ij}) \\ \tau_{ij} &= a_{ij} + \frac{b_{ij}}{T} + e_{ij} \ln T + f_{ij} T \\ \alpha_{ij} &= c_{ij} + d_{ij} (T - 273.15 K) \\ \tau_{ii} &= 0 \\ G_{ii} &= 1 \end{aligned} \quad (2)$$

Assuming a null value for d_{ij} , d_{ji} , e_{ij} , e_{ji} , f_{ij} and f_{ji} , the fitting of the binary interaction parameters (a_{ij} , a_{ji} , b_{ij} , b_{ji}) and of the nonrandomness parameter, $\alpha_{ij} = \alpha_{ji}$, has been performed with the Data Regression System of Aspen Plus® V14.0, selecting the ordinary least square objective function, the Britt and Luecke as main algorithm and the Deming method for its initialization.

Table 4

Data of temperature, pressure, composition of the liquid and of the vapor phases collected at PT lab of Politecnico di Milano at a pressure of 60 kPa with the related average expanded uncertainties (U), (x is the mole fraction in the liquid phase and y is the mole fraction in the vapor phase of the most volatile component, *i.e.*, 2-Propanol).

P [kPa]	T [K]	x	y
60	342.85	1.0000	1.0000
60	342.00	0.8986	0.8922
60	341.14	0.7865	0.7613
60	340.94	0.7140	0.6898
60	340.92	0.6461	0.6492
60	341.05	0.5895	0.5998
60	341.25	0.4943	0.5690
60	341.42	0.4139	0.5347
60	341.88	0.3452	0.5250
60	342.02	0.2951	0.5056
60	342.92	0.1401	0.4769
60	344.12	0.0902	0.4638
60	346.57	0.0690	0.3998
60	348.30	0.0371	0.3473
60	350.29	0.0323	0.2924
60	352.72	0.0170	0.2499
60	354.75	0.0145	0.1655
60	356.52	0.0110	0.1327
60	358.68	0.0063	0.0418
60	359.01	0.0000	0.0000

The average expanded uncertainties are $U(\text{pressure}) = 0.5$ kPa, $U(\text{temperature}) = 0.073$ K, $U(\text{liquid mole fraction of 2-Propanol}) = 0.0215$ mol/mol, $U(\text{vapor mole fraction of 2-Propanol}) = 0.0278$ mol/mol.

Table 5

Data of temperature, pressure, composition of the liquid and of the vapor phases collected at PT lab of Politecnico di Milano at a pressure of 80 kPa with the related average expanded uncertainties (U), (x is the mole fraction in the liquid phase and y is the mole fraction in the vapor phase of the most volatile component, *i.e.*, 2-Propanol).

P [kPa]	T [K]	x	y
80	349.57	1.0000	1.0000
80	349.45	0.9848	0.9787
80	349.26	0.9709	0.9578
80	349.01	0.9504	0.9286
80	348.44	0.8929	0.8607
80	347.83	0.8069	0.7683
80	347.61	0.7055	0.6948
80	347.60	0.6757	0.6784
80	347.62	0.6540	0.6641
80	347.66	0.6064	0.6422
80	347.83	0.5141	0.6067
80	348.42	0.3835	0.5797
80	348.80	0.2795	0.5662
80	349.36	0.1719	0.5476
80	350.68	0.0941	0.4943
80	353.49	0.0441	0.4294
80	356.69	0.0209	0.3541
80	359.73	0.0153	0.2527
80	362.78	0.0095	0.1685
80	366.57	0.0000	0.0000

The average expanded uncertainties are $U(\text{pressure}) = 0.5$ kPa, $U(\text{temperature}) = 0.036$ K, $U(\text{liquid mole fraction of 2-Propanol}) = 0.0300$ mol/mol, $U(\text{vapor mole fraction of 2-Propanol}) = 0.0286$ mol/mol.

The regressed parameters obtained in the first regression reported a value of $\alpha_{12} = \alpha_{21}$ equal to 0.43, a value close to 0.47, that is the one suggested for the nonrandomness parameter by Renon and Prausnitz [37] for Type VII mixtures. According to the authors of the NRTL theory, this type of mixtures include systems of Water and a polar self-associated substance and are characterized by a high degree of nonrandomness. Previous studies in the literature [80,81] focused on the analysis of the IPA + Water system with Raman spectra to better understand the molecular association for this binary system. Idrissi and Longelin [80], in

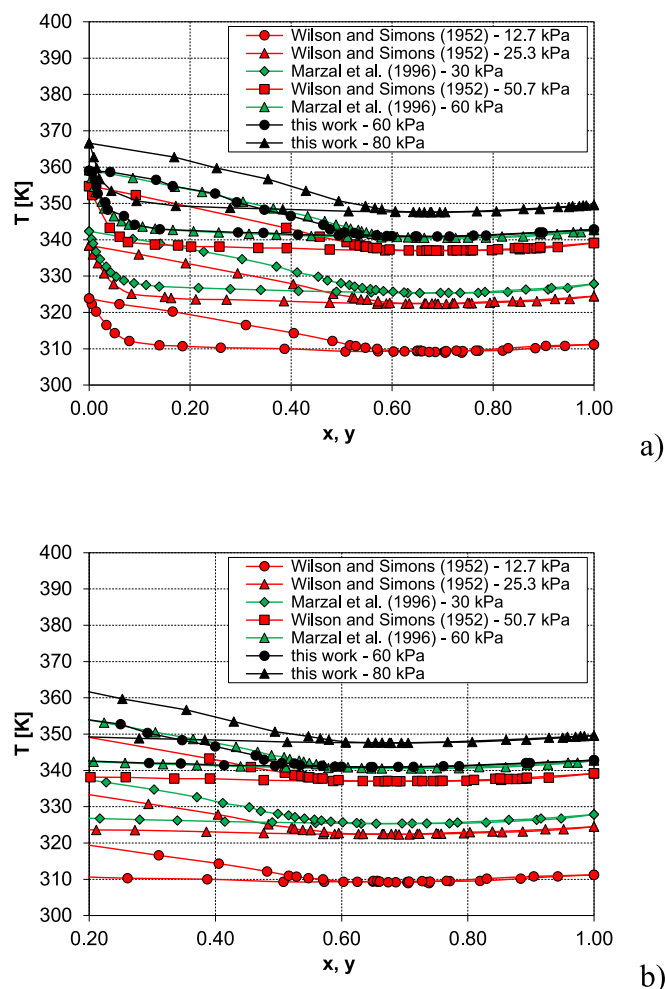


Fig. 6. Experimental VLE data collected at PT lab of Politecnico di Milano and taken from the literature for the 2-Propanol + Water system at pressures below 100 kPa a) for a mole fraction of IPA ranging from 0 to 1 and b) for a mole fraction of IPA close to the azeotropic point.

Table 6

Results of the thermodynamic consistency test performed for the experimental points collected at PT lab in this work.

Data set	Value	Tolerance	Result
P = 60 kPa	-4.83377669	10 %	Passed
P = 80 kPa	-0.654151307	10 %	Passed

particular, reported that IPA has a high tendency to self-aggregate, therefore the IPA + Water system can be considered a Type VII mixture. The regression has, then, focused on the determination of the values of the binary interaction parameters a_{ij} , a_{ji} , b_{ij} , b_{ji} with a fixed value of the nonrandomness parameter equal to 0.47. The obtained values are listed in Table 7.

The percent average absolute deviation (AAD%) for all the predicted variables has been calculated, according to Eq. (3), where n is the

Table 7

Regressed NRTL binary interaction parameters and the nonrandomness parameter for the 2-Propanol(1) + Water(2) system.

α_{12}	α_{21}	b_{12} [1/K]	b_{21} [1/K]	$\alpha_{12} = \alpha_{21}$
-1.39470283	4.50009825	627.160288	-791.079743	0.47 (not regressed)

number of experimental points for each set of data, in order to assess the ability of each model to fit the experimental data. Table S1, Table S2 and Table S3 report the AAD% values for the isobaric cases, Table S4, Table S5 and Table S6 contain the AAD% of the isothermal data sets. These latter tables can be found in the Supporting information file.

$$AAD\% = \frac{100}{n} \sum_{i=1}^n \frac{|calc\ value_i - exp\ value_i|}{exp\ value_i} \quad (3)$$

In Table 8 the averages of all the AAD% values for each calculated variable are reported. The NRTL model with regressed parameters is characterized by lower AAD% values than the NRTL (default parameters) and similar to the one provided by the UNIFAC (Dortmund modified), for the isobaric datasets. The absolute deviations obtained for the NRTL model with the binary interaction parameters reported in Table 7 are detailed in Table 9. The deviation of the pressure for the isothermal datasets is influenced by the deviation in estimating the experimental data by Shimoyama et al. [74] at pressures up to 11480 kPa, with an AAD% ranging from 5.02 % to 15.36 %.

During the regression the isothermal set of data collected by Barr-David and Dodge [73] at 200 °C failed the thermodynamic consistency test (area test of Redlich-Kister). Hence, it has not been considered in the calculation of the regressed parameters.

Fig. 7 reports the representation of the boiling point curve and of the dew point curve obtained with the NRTL method with the binary interaction parameters determined in this work (Table 7) and the experimental points for the comparison. The estimated azeotropic point is in agreement with the experimental data.

The non-ideality of the system can be determined also by analyzing the activity coefficients of the two components of interest in the binary mixture IPA + Water. Fig. 8 reports the experimental activity coefficient, determined on the basis of the experimental data reported in Table 4 and in Table 5, and the values obtained by employing the NRTL model with the binary interaction parameters determined in this work (Table 7). The activity coefficient increases significantly for the component in the binary mixture, with values up to more than 13 for 2-Propanol and more than 3 for Water, and tends to the unitary value for the pure components ($x = 1$), as theoretically expected.

5. Conclusions

This work reports isobaric experimental Vapor-Liquid Equilibrium data for the system 2-Propanol + Water at 60 kPa and at 80 kPa collected at PT lab of Politecnico di Milano, to address the lack of experimental data present in the literature under vacuum conditions. The system has been described considering different thermodynamic models (NRTL,

Table 8

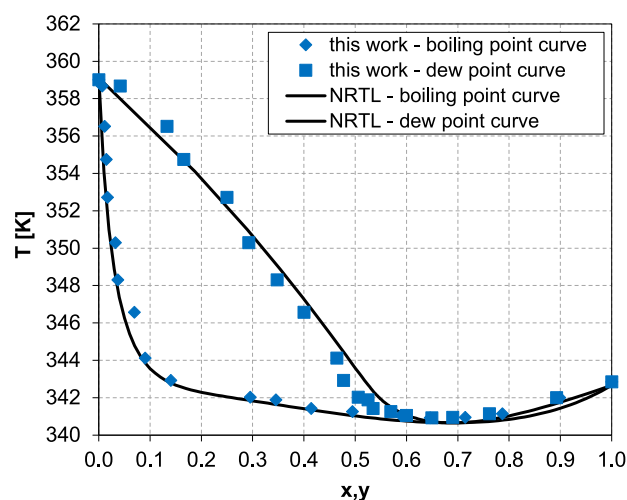
Average of the AAD% for each calculated variable considering all the thermodynamic models under study.

Isobaric datasets			
	AAD%(T)	AAD%(y 2-Propanol)	AAD%(y water)
PSRK	0.2226	6.682	5.815
NRTL (default parameters)	0.1388	7.943	4.413
UNIFAC (Dortmund modified)	0.1333	5.559	3.989
Wilson	0.1898	7.831	5.473
NRTL (regressed parameters)	0.1359	6.366	4.099
Isothermal datasets			
	AAD% (P)	AAD% (y 2-Propanol)	AAD% (y water)
PSRK	2.087	5.433	4.612
NRTL (default parameters)	9.172	33.61	9.314
UNIFAC (Dortmund modified)	7.026	12.15	5.816
Wilson	9.373	32.89	9.831
NRTL (regressed parameters)	3.382	16.470	4.790

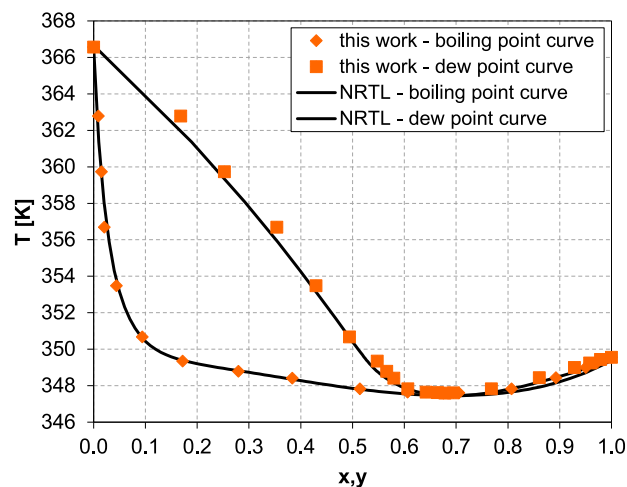
Table 9

Average absolute deviations for each calculated variable considering the NRTL thermodynamic model with the binary interaction parameters reported in Table 7.

	Absolute deviation on T [K]	Absolute deviation on P [kPa]	Absolute deviation on y _{2-Propanol}	Absolute deviation on y _{water}
Isobaric datasets	0.29207	n.a.	0.01980	0.01462
Isothermal datasets	n.a.	123.367	0.02585	0.02585



a)



b)

Fig. 7. Experimental VLE data collected at PT lab of Politecnico di Milano and obtained with the NRTL model with the binary interaction parameters determined in this work for the 2-Propanol + Water system for a) 60 kPa and b) 80 kPa. x, y refer to the mole fraction of IPA in the liquid and in the vapor phase.

UNIFAC (Dortmund modified), Wilson and PSRK) and, then, the regression of the binary interaction parameters has been accomplished for the NRTL model. Finally, for each thermodynamic model the AAD% has been calculated. For the isobaric datasets the NRTL (regressed parameter) and the UNIFAC (Dortmund modified) show the best fit, while for the isothermal datasets the PSRK turns out to be better. The regression of the NRTL binary interaction parameters lowers the AAD% obtained by employing this thermodynamic model, for both the isobaric

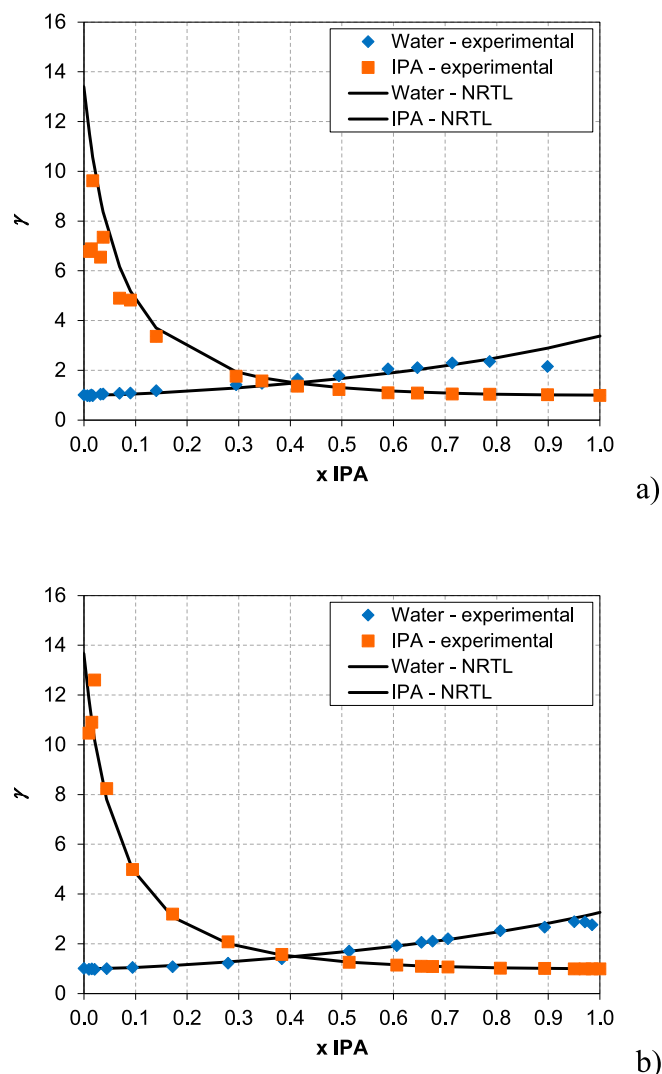


Fig. 8. Values of the activity coefficients for IPA and Water as the mole fraction of IPA in the liquid phase varies for isobaric data collected at PT lab in this work a) at 60 kPa and b) at 80 kPa. The activity coefficients have been determined from the experimental points (symbols) and by using the NRTL model (lines) with the binary interaction parameters determined in this work (Table 7).

and the isothermal sets of experimental data.

CRediT authorship contribution statement

Camilla Barbieri: Writing – original draft, Validation, Investigation, Data curation. **Giorgia De Guido:** Writing – review & editing, Conceptualization. **Stefania Moiola:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Percent Absolute Average Deviation (AAD%) for all the predicted variables for each thermodynamic model which has been considered in this study for both the isobaric and isothermal sets of data. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jct.2024.107342>.

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